

DWDM Ring-Resonator Circuit Design

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The main circuit designed in this project is a compact wavelength-selective DWDM structure based on two silicon microring resonators. The circuit was implemented using the UBC EBeam PDK in KLayout and was designed to be compatible with the automated Phot1x measurement setup. Four TE grating couplers are arranged vertically with the required 127 μm pitch. The second grating coupler from the top is used as the optical input, while the remaining three couplers provide two wavelength-selective drop outputs and one through output.

The optical signal enters the circuit through port **B** and propagates along the main bus waveguide toward output **D**. Two add-drop microring resonators are coupled to this bus. One ring directs its resonant wavelength toward output **A**, while the other directs its resonant wavelength toward output **C**. Wavelengths that are not efficiently coupled into either ring remain in the main bus and exit through output **D**. The circuit therefore behaves as a simple wavelength demultiplexer:

$$B \rightarrow \begin{cases} A & \text{Ring A resonance} \\ C & \text{Ring C resonance} \\ D & \text{off resonance} \end{cases}$$

The rings were designed using a 500 nm silicon waveguide width and a coupling gap of approximately 200 nm. The two rings use slightly different radii, approximately 6.8 μm and 6.9 μm , so that their resonance spectra are shifted relative to each other. The intention is to obtain two different wavelength channels around the 1550 nm telecommunications band, approximately corresponding to 1540 nm and 1560 nm.

The free spectral range of a microring can be approximated by:

$$FSR \approx \frac{\lambda^2}{n_g 2\pi R}$$

Where R is the ring radius and n_g is the waveguide group index. Taking $n_g \approx 4.2$ at 1550 nm gives an expected FSR of approximately 13.4 nm for the 6.8 μm ring and 13.2 nm for the 6.9 μm ring. Therefore, each output should show a periodic series of resonances separated by roughly 13 nm. At the corresponding resonance, outputs A or C should show a transmission peak, while the ‘through’ port D should show a complementary transmission dip.

The exact resonance wavelengths and linewidths cannot be predicted from radius alone because they also depend on the fabricated effective index, propagation loss, and coupling

strength. Assuming a representative finesse of approximately 20, the expected resonance linewidth would be on the order of 0.65-0.7 nm. The 200 nm coupling gap strongly influences the loaded Q-factor and therefore the measured extinction ratio and linewidth.

In addition to the DWDM structure, the layout contains simple reference structures, including a direct waveguide reference and a single-ring reference. These provide baseline measurements for grating-coupler/waveguide loss and for comparison with the more complex two-ring DWDM circuit.

Overall, the design is intended to demonstrate wavelength-selective routing using compact silicon microring resonators while remaining compatible with the fabrication rules, chip-area constraints, and automated fiber-array measurement system of the Phot1x course.